

ST AUGUSTINE AP, FL, USA

WMO: 722212

Lat: 29.959N

Lon: 81.340W

Elev: 10

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 92814

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	34.1	37.3	18.0	13.6	42.4	23.1	17.4	44.2	24.7	54.4	22.0	53.8	6.6	310	0.402

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.1	92.4	77.7	90.5	77.8	88.7	77.7	80.6	87.8	79.9	87.2	79.1	86.2	9.4	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.8	149.4	85.1	77.6	143.4	84.3	77.0	140.4	83.7	44.1	88.1	43.2	87.2	42.5	86.2	86.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
21.1	18.8	17.0	DB	27.3	96.0	2.7	2.4	25.3	97.7	23.7	99.0	22.1	100.4	20.1	102.1
			WB	24.6	82.3	3.4	1.3	22.2	83.2	20.1	84.0	18.2	84.7	15.7	85.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	70.1	55.7	58.9	63.0	68.9	75.0	80.0	81.5	81.5	79.4	72.9	64.8	59.1	(6)
	DBStd	11.03	9.10	8.11	7.56	5.57	4.34	2.90	2.31	2.24	2.70	5.98	7.41	8.84	(7)
	HDD50	97	51	16	6	0	0	0	0	0	0	0	3	21	(8)
	HDD65	993	307	200	126	27	2	0	0	0	0	13	93	225	(9)
	CDD50	7437	227	266	409	566	776	899	976	975	883	711	447	302	(10)
	CDD65	2858	18	30	64	143	312	449	511	510	433	260	87	41	(11)
	CDH74	23746	61	121	295	748	2284	4048	5004	5045	3951	1747	317	125	(12)
	CDH80	7268	3	12	43	118	573	1404	1897	1794	1099	288	30	7	(13)
(14) Wind	WSAvg	8.2	7.8	8.0	8.6	8.9	8.9	7.8	7.1	7.2	8.7	8.8	8.4	7.7	(14)
(15) Precipitation	PrecAvg	50.9	2.8	2.8	3.4	2.5	3.0	6.9	5.4	6.4	8.6	4.8	2.2	2.5	(15)
	PrecMax	66.2	7.9	8.5	8.9	8.8	7.1	11.2	11.5	12.9	17.0	12.0	5.7	10.1	(16)
	PrecMin	32.9	0.9	0.6	0.2	0.6	1.0	0.7	1.7	2.9	2.0	0.3	0.2	0.4	(17)
	PrecStd	8.9	1.9	2.0	2.4	2.1	1.9	2.6	2.3	2.6	4.4	3.1	1.4	2.5	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.5	81.8	84.5	87.6	91.9	95.8	95.1	93.7	91.0	88.0	83.0	81.0	(19)
		MCWB	67.8	68.3	68.6	70.7	73.3	77.2	78.3	78.4	76.3	76.9	73.8	72.1	(20)
	2%	DB	75.1	77.8	80.7	83.0	88.2	91.9	91.6	91.0	88.4	84.7	80.5	77.9	(21)
		MCWB	66.7	67.8	67.0	70.4	72.9	77.2	78.3	79.1	77.0	75.7	72.3	70.3	(22)
	5%	DB	72.2	74.2	77.3	80.8	84.9	89.4	90.1	89.6	86.8	83.0	77.5	74.0	(23)
		MCWB	65.1	65.7	66.6	69.4	73.4	77.4	78.4	79.1	77.0	74.5	70.6	68.6	(24)
	10%	DB	69.1	71.3	74.3	78.6	82.6	87.1	88.2	87.9	85.3	81.4	75.2	71.7	(25)
		MCWB	63.7	64.6	65.1	68.6	72.6	76.9	78.2	78.6	76.6	73.3	69.0	67.5	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.9	71.3	72.6	75.1	77.5	80.6	81.5	81.9	80.5	79.1	76.5	73.8	(27)
		MCDB	75.8	77.2	78.8	81.8	85.2	88.8	89.0	88.9	86.4	84.5	80.7	78.5	(28)
	2%	WB	68.6	69.5	70.2	73.4	76.2	79.4	80.4	80.8	79.2	77.5	74.6	71.7	(29)
		MCDB	73.3	75.0	76.7	79.6	83.3	88.1	88.1	87.8	85.3	82.8	78.4	75.9	(30)
	5%	WB	66.6	67.6	68.8	72.0	75.2	78.3	79.7	80.0	78.3	76.3	72.3	69.8	(31)
		MCDB	71.1	72.5	74.9	77.9	81.9	86.3	87.5	87.0	84.3	81.3	76.0	73.2	(32)
	10%	WB	64.3	65.4	67.1	70.4	74.2	77.5	78.7	79.2	77.6	75.2	70.2	67.8	(33)
		MCDB	67.8	69.7	72.4	76.2	80.8	84.9	86.2	86.0	83.5	80.0	74.3	71.0	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	17.6	16.6	17.0	16.3	15.0	14.0	14.4	13.1	11.6	13.7	15.6	16.4	(35)
		MCDBR	18.7	19.2	19.5	18.0	18.0	17.9	16.7	15.4	13.6	13.1	15.3	16.6	(36)
		MCWBR	12.2	11.0	10.4	9.2	7.6	7.4	7.3	6.9	6.1	6.7	9.0	10.7	(37)
	5% WB	MCDBR	16.7	16.9	16.6	15.4	14.6	15.2	14.9	13.8	12.0	11.4	12.8	15.1	(38)
		MCWBR	11.6	10.8	10.2	8.7	7.2	7.3	7.5	7.1	6.1	6.1	8.7	10.2	(39)
(40) Clear-Sky Solar Irradiance	taub		0.339	0.352	0.371	0.396	0.433	0.472	0.497	0.482	0.427	0.400	0.362	0.346	(40)
	taud		2.540	2.513	2.461	2.375	2.314	2.270	2.203	2.252	2.401	2.444	2.498	2.542	(41)
	Ebn at Noon		284	290	290	285	273	261	254	257	269	270	273	276	(42)
	Edh at Noon		28	31	35	39	42	43	46	43	36	33	29	26	(43)
(44) All-Sky Solar Radiation	RadAvg		1002	1204	1572	1918	2036	1896	1880	1761	1521	1343	1085	910	(44)
	RadStd		74	124	137	115	128	145	117	115	124	110	63	78	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (33 neighbors)	+0.79	+1.39	N/A	N/A	N/A	N/A	-16	-112	+288	+209

Nomenclature: See separate page